# MEASURING THE FRACTAL PROPERTIES OF THE SIERPIŃSKI TRIANGLE

Educational paper on the properties of the Sierpiński Triangle, including the measurement of the area and perimeter.

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# 1 Introduction

This educational mathematical study was written to inspire and enlighten an audience that may not be familiar yet with the beauty of the Sierpiński Triangle fractal. This thesis focuses on the understanding of the processes behind all the fascinating calculations. All the claims in this paper are derived from simple facts that can be deduced about this shape.

I hope this study will help you develop another perspective on fractals and how to compute their properties. Now, onto the shape itself.

The Sierpiński triangle is a fractal constructed by an infinite series of smaller triangles that fit together in a pattern. A whole triangle is divided into four smaller ones, with the middle one not belonging to the whole shape. For every smaller triangle created, this process is infinitely repeated; therefore, continuously smaller triangles are being subtracted from the whole shape.

# 2 Fundamental values

## 2.1 Fractal dimension

$$R^D = N$$
$$D = \log_R(N) \tag{1}$$

## 2.2 Feasibility of computing fractal properties

Before trying to calculate other properties of this fractal, we must know if it is even possible to get a glimpse of its properties represented as finite values. To answer this question, first, we must tear this shape apart to its core. This shape seems complex, but it is in reality constructed only from small triangles. This fact alone should nudge us in the right direction.

**Axiom 1** *If we can tear apart a shape to make it simpler with each iteration, it should be possible to compute the area and the perimeter of this shape.*

The properties of this triangular fractal can be easily calculated and can also be logically derived from the facts that we have gathered about this shape. The compounded area of the Sierpiński triangle must have a finite value since it is being stored within a 2D space that is not altering its size at any given point. Hence, the whole area is a finite number, despite the fact that the process through which we can obtain this value is infinite. Similarly, the perimeter is also stored in a 2D, non-altering space, but with one major difference. The perimeter gets longer with every iteration, unlike the area, which gets smaller due to the triangle division at every iteration.

# 3 Area computation at $n$th iteration

Before trying to compute the area at any given iteration, we first need to understand the process by which this value is constructed. This process, like previously said, is an infinite series of the same operations but executed on smaller and smaller parts of the fractal. Thus, we must reckon with all the possible $n$ iterations from 0 to infinity.

## 3.1 First approach: Area reduction

The main triangle will get divided into 4 equal parts, with exactly 1 triangle being removed from the total area. You can notice that at every iteration, the area is always reduced by a $\frac{1}{4}$th of the triangle's area; therefore, the area is always $\frac{3}{4}$ of the previous triangle's area. Now, we can model an equation that is going to express the relationship between $n$ and the area.

The area at each step takes $\frac{3}{4}$ and multiplies it by itself $n$ times. So for notational simplicity, we can raise it to the $n$th power. To also count in the initial triangle's area, we can plug in $A_0$ and multiply it by the expression:

$$A_n = \left(\frac{3}{4}\right)^n \cdot A_0 \tag{2}$$

where: $A_n$ = the computed area at $n$th iteration;

$A_0$ = the initial triangle's area;

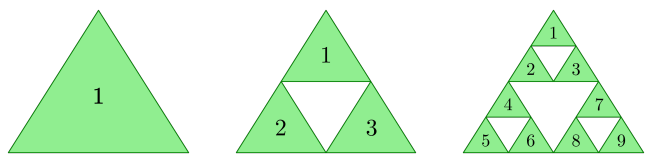
$n$ = current iteration, $n \in \mathbb{N}$.

## 3.2 Second approach: Counting triangles

We can also use an alternative approach that will lead us to the same conclusion. This method can be more intuitive or more comprehensive for the audience since we are working with more arbitrary values than we were with the previous method. In the last process, we tried reducing the area each iteration by a $\frac{1}{4}$th of the triangle's area.

In the second approach, we will try adding up all the small triangles' areas

that belong to the shape. You can notice that the number of triangles inside the shape is always increased by a factor of 3. This means that the total number of triangles at $n$th iteration is always $3^n$.




(a) $3^0$ triangles (b) $3^1$ triangles (c) $3^2$ triangles

Figure 1: Three simple graphs

Once we have established the number of triangles, we need to compute the area that each triangle is going to take up. The whole area of a triangle at $n$th iteration is always divided into four equal parts. So, we can say that the area of one of those 4 triangles is equal to $\frac{1}{4}$ of its area.

Depending on the current iteration, the partial area will get smaller when the iteration property gets higher. At every iteration, the area of the 3 remaining triangles is getting smaller by a $\frac{1}{4}$th. This means that the area of the small triangles is $\frac{1}{4^n}$ multiplied by the initial triangle's area.

We can now simply combine those two facts into an equation. If we know that every $n$th iteration has $3^n$ triangles and that every triangle has an area of $\frac{1}{4^n}$, that means that the area at any given iteration is:

$$A_n = \frac{3^n}{4^n} \cdot A_0 = \left(\frac{3}{4}\right)^n \cdot A_0 \tag{3}$$

As you can see, we have derived the same equation like we did in equation (2). This means that no matter which route we choose to take, the result remains the same.

### 3.2.1 Example

We have previously explored how to derive the formula for the computation of the area at the nth iteration. Now, we will take a look at a basic example of how to use this formula. For example, let's say that we want to compute the

area of this fractal at the 3rd iteration, with the initial triangle's area being 10. We need to make use of the formula we came up with earlier:

$$\begin{aligned} A_{10} &= \left(\frac{3}{4}\right)^n \cdot A_0 = \left(\frac{3}{4}\right)^3 \cdot 10 \\ &\approx 4.22 \end{aligned} \tag{4}$$

### 3.3 Third approach: Visual representation

There exists a third way to compute the area at the $n$th iteration of the Sierpiński triangle; it is also the last alternative that we are going to cover in this paper. This alternative is more visual than the others, and its core depends on the rearrangement of the triangles that make up the whole fractal.

First and foremost, we need to lay next to each other and on top of each other all the triangles that are contained in a certain iteration. The first triangle is going to sit with its base down, and next to it will sit the second triangle with its base up. This process is repeated until we have run out of available triangles to lay in. We can visualize the second iteration like this:

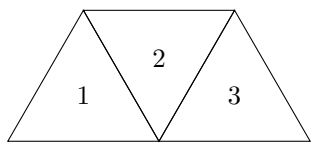



Figure 2: Rearrangement of triangles

Now, when we have laid down all the triangles that are available at the $n$th iteration, we can start rearranging the triangles into a new shape that can be more easily calculated. Take the last triangle and split it by the height of the equilateral triangle into two equal parts. Leave the first half of the triangle at the end of the series, but move the second half to the very beginning and flip it by the horizontal axis.

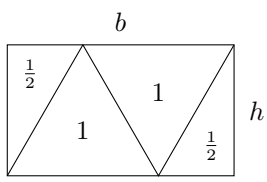



Figure 3: Triangles in a rectangle pattern

Notice that the area of the new shape did not change and is still equal to the previous series of laid-down triangles. One thing that did change, however, was the shape. The new shape is a rectangle created by the triangle series and can now be more easily visualized. We can express the relationship between the fractal area and the rectangle's properties in the following way:

$$A_n = b_n \cdot h_n \tag{5}$$

Two variables that still remain to be figured out are the base and the height of the new rectangle. Let's start with the base of the rectangle. You can notice that when all the triangles are laid down to fit in the rectangle, half of them have their bases down and half of them have their bases up. This means the length of the rectangle's base must equal the number of triangles in the desired iteration divided by 2, therefore $\frac{3^n}{2}$.

We must also take into account the side length of the triangles that make up the whole fractal; we can call the zeroth iteration's triangle side length $a$. We know that the side length of the first iteration's triangle is $\frac{1}{2}a$, so if we were to get the ratio between them, we would see that the side lengths keep getting smaller by the factor of $\frac{1}{2}$. This means that the side length of a triangle at $n$th iteration is equal to $a \cdot \frac{1}{2^n}$ or $\frac{a}{2^n}$.

Now, we just have to multiply the number of the triangle bases that make up the whole fractal with the size of the triangle's side lengths:

$$b_n = \left(\frac{3^n}{2}\right)\left(\frac{a}{2^n}\right) = \frac{3^n \cdot a}{2^{n+1}} \tag{6}$$

Once we have established the base of the rectangle, we can move on to the rectangle's height. You can notice that the rectangle's height is the same as the height of the triangles that are present in the rectangle. So first, we need to determine the height of the equilateral triangles that make up the fractal. From a short calculation, we can say that a triangle with the side length $a$ has the height $\frac{\sqrt{3}}{2}a$.

$$\begin{aligned} A_n &= \left(\frac{3^n}{2}\right)\left(\frac{a}{2^n}\right) \cdot \left(\frac{\sqrt{3}}{2}\right)\left(\frac{a}{2^n}\right) \\ &= \frac{3^n \cdot a^2\sqrt{3}}{2^{2n+2}} \end{aligned} \tag{7}$$

# 4 Perimeter computation at $n$th iteration

Before we can get started, we first need to take into account the properties on which this computation depends. In order to compute this, we need to find out the total number of individual triangles that together compose the whole fractal and the lengths of the triangles.

First, we are going to compute the total number of triangles that make up the fractal. You can notice that with each iteration, the total number of triangles is multiplied by a factor of 3. This means that the total number of triangles at any given iteration is $3^n$. Once the total number of triangles is settled, we have to calculate the total number of sides that the triangles together contain. This means taking the total number of triangles at $n$th iteration and multiplying it by a factor of 3, since each triangle has 3 sides; therefore, the total number of triangle sides is $3^{n+1}$.

$$P_n = \frac{3^{n+1}}{2^n} \cdot a \tag{8}$$

# 5 Approaching the limits of computation

We can now compute the precise area and perimeter properties at a certain iteration. Those values can help us when we are reckoning with finite $n$th iteration's values, but when we take into account the infinite side of things, those equations will not be able to provide us the solutions we desire anymore.

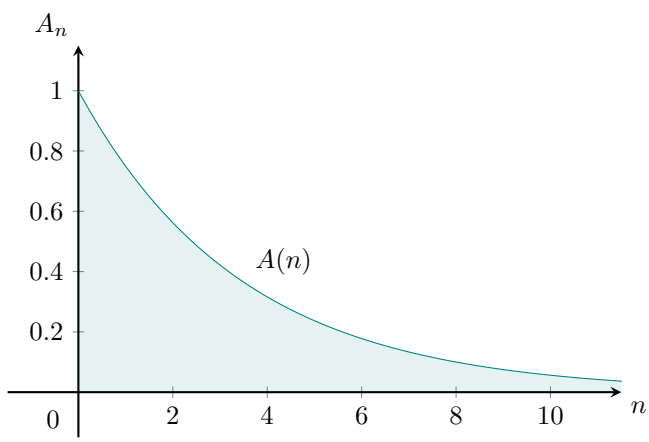
This is the moment when we need to introduce the limits of these computational functions. Testing the area and perimeter functions will help up solve these properties in an infinite space. We will be able to approach the limits by simulating extreme inputs to these functions.

## 5.1 Approaching the area limit

We have previously looked into the process of computing the area, from which we have received the rule for the calculation. Now, we can get the gist of that function, plug it into the limit $L_a$ and see to what extent it approaches. The gist of the area function is equal to $\left(\frac{3}{4}\right)^n$. Now we need to plug it into the limit with respect to the variable $n$ going to infinity:

$$L_a = \lim_{n\to\infty} \left(\frac{3}{4}\right)^n \tag{9}$$

Notice that no matter how high the variable $n$ goes in the examined limit, the function is not increasing in value. So, we can determine that the function is monotonically non-increasing, meaning that the area will get infinitely smaller with each iteration. We can depict this in a graph $A(n) = \left(\frac{3}{4}\right)^n$ and see how the variable $A_n$ will respond over time.




(10)

The constructed graph, with respect to both $A_n$ and $n$, only agrees that the area will get infinitely smaller and will continually get closer to 0. There is another way to prove this: notice that the denominator approaches infinity quicker than the numerator since $4^n > 3^n$; meaning that the whole fraction will keep getting smaller in value. Now, we can say for certain that the limit of the function will approach the value of 0.

$$L_a = 0 \tag{11}$$

## 5.2 Approaching the perimeter limit

$$\lim_{n\to\infty}\left(\frac{3^{n+1}}{2^n}\right)=\infty \tag{12}$$